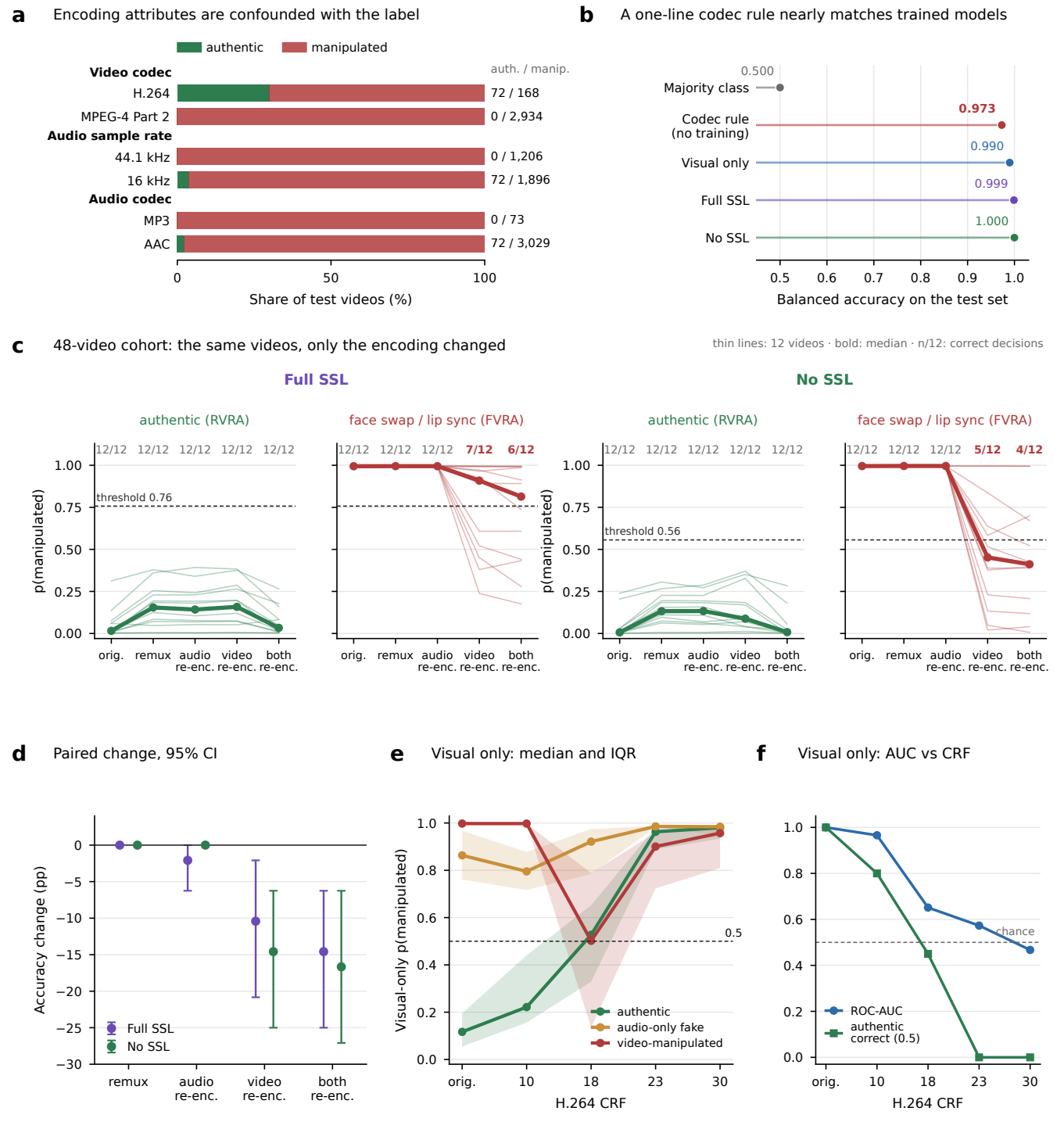
a
Encoding attributes are confounded with the label
authentic
manipulated
auth. / manip.
Video codec
H.264
72 / 168
MPEG-4 Part 2
0 / 2,934
Audio sample rate
44.1 kHz
0 / 1,206
16 kHz
72 / 1,896
Audio codec
MP3
0 / 73
AAC
72 / 3,029
0
50
100
Share of test videos (%)
b
A one-line codec rule nearly matches trained models
0.500
Majority class
0.973
Codec rule
(no training)
0.990
Visual only
0.999
Full SSL
1.000
No SSL
0.5
0.6
0.7
0.8
0.9
1.0
Balanced accuracy on the test set
c
48-video cohort: the same videos, only the encoding changed
thin lines: 12 videos · bold: median · n/12: correct decisions
Full SSL
No SSL
authentic (RVRA)
face swap / lip sync (FVRA)
12/12
7/12
6/12
5/12
4/12
threshold 0.76
threshold 0.56
p(manipulated)
1.00
0.75
0.50
0.25
0.00
orig.
remux
audio re-enc.
video re-enc.
both re-enc.
d
Paired change, 95% CI
Accuracy change (pp)
0
−5
−10
−15
−20
−25
−30
Full SSL
No SSL
remux
audio re-enc.
video re-enc.
both re-enc.
e
Visual only: median and IQR
Visual-only p(manipulated)
1.0
0.8
0.6
0.4
0.2
0.0
0.5
authentic
audio-only fake
video-manipulated
orig.
10
18
23
30
H.264 CRF
f
Visual only: AUC vs CRF
chance
ROC-AUC
authentic correct (0.5)
H.264 CRF